Calculus I　　微积分(1)

Instructor

Assoc. Prof. Peng JIN 金鹏

Office: T3-602-R4 Tel:3620406

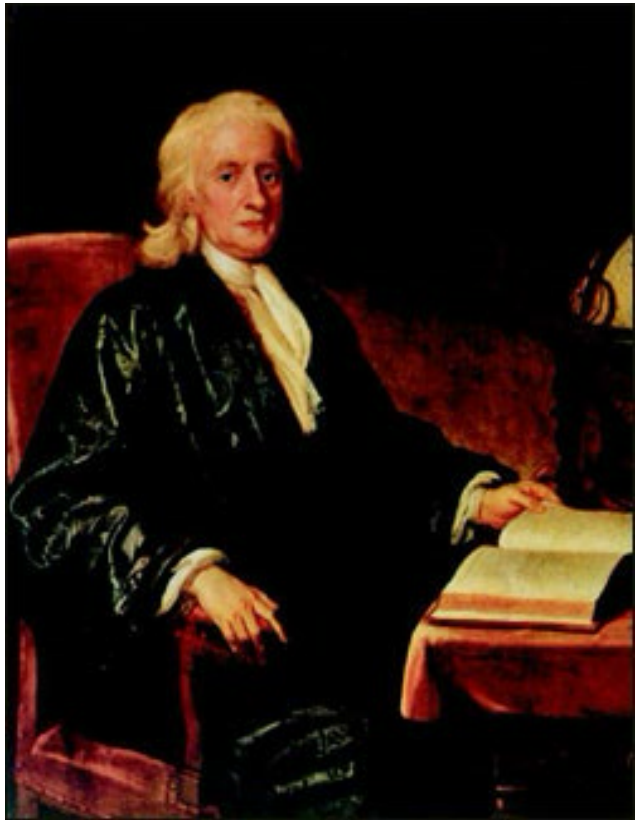
Email: pengjin@uic.edu.cn

T.A.: Mr. Qianzi CAO

Office: T3-602-R25-H6

Email: qianzicao@uic.edu.cn

They invented Calculus!



Sir Isaac Newton
(1642-1727)



Gottfriend Wilhelm von Leibniz

(1646-1716)

Calculus was continued….



Fermat(费马)

(1601－1665)



Euler(欧拉)

(1707－1783)



Gauss(高斯)

(1777－1855)



Cauchy(柯西)

(1789－1857)

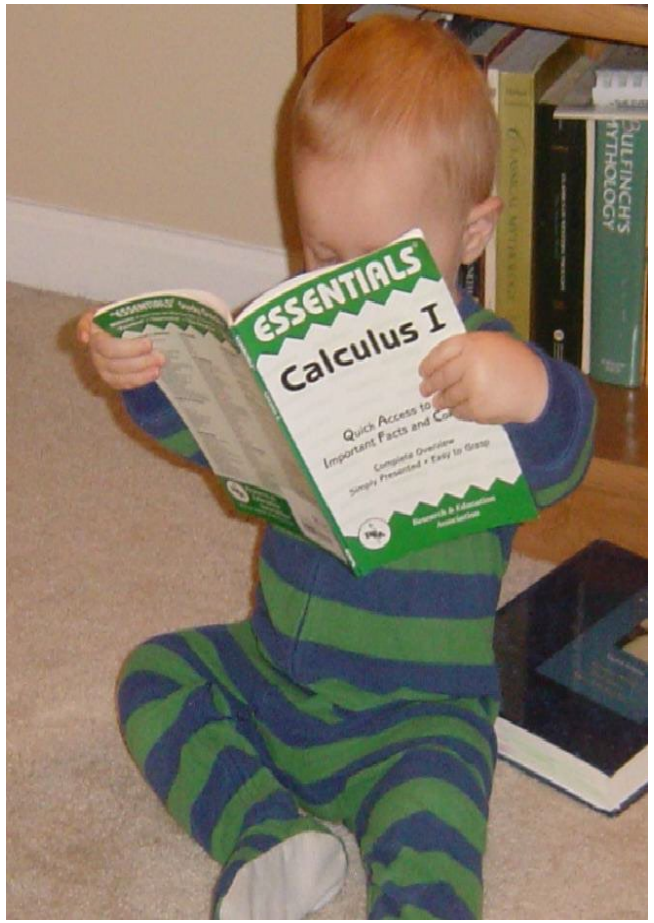
What is Calculus all about?

- Calculus is the study of change, or to be more precise, changing quantities.

- The two key areas of Calculus are Differential Calculus and Integral Calculus.

- The big surprise is that these two seemingly unrelated areas are actually connected via the Fundamental Theorem of Calculus.

Why do I have to take Calculus?



- If you choose a career path that requires any level of mathematical sophistication (such as Engineering, Economics or any branch of Science), you will need Calculus.
- Emphasize on proofs and conceptual understanding rather than computation.
- To succeed in Calculus (and Mathematics in general) you must be able to: solve problems, reason logically and understand abstract concepts. The ability to master these skills is a key indicator of success in future academic endeavors.

What can I do to maximize my chances for success?

1. Understand, don't memorize.
2. Ask why, not how.
3. More practices and exercises.
4. Learn techniques, not results.
5. Make sure you understand each topic before going on to the next.

Some Suggestions

- Bring your questions when you take lecture. Questions could be found by preview.

- Review what you have studied as soon as possible after lecture. Textbook should be read carefully.

- Try your best to attend all lectures and tutorial class.

Assessment Method

- **Class Participation** 5%
- **Assignments** 15%
  - Problem sets are assigned weekly, and due the following week
- **2 Quizzes** 20%
  - Quiz 1: Sep 21, 9:00-10:00, relevant: Chap 1
  - Quiz 2: Nov 23, 9:00-10:00, relevant: Chap 4
- **Mid-term Test** 20%
  - Oct 26, 9:00-11:00, relevant: Chap 1-Chap 3
- **Final examination** 40%

CRA (Criterion-Referenced Assessment)

Adoption of the Criterion-Referenced Assessment (CRA) for evaluating students' performance

Outcome-Based Teaching and Learning (OBTL)

CRA model is directly compatible with the OBTL philosophy.

CR Assessment – Assignment/Quiz/Midterm test/

Criteria for assessment	*Performance levels*				
	Excellent 10/9	**Good 8/7/6**	**Satisfactory 5/4**	**Marginal Pass 3/2/1**	**Fail 0**
Correctness (80 % weighting)	Excellent understanding of and response to the questions. The steps in the proof / computation are complete and correct.	Good understanding of and response to the questions. The steps in the proof / computation are mostly correct, but involve some minor flaws.	Satisfactory understanding of and response to the questions. The steps in the proof / computation are correct but involve some moderate flaws.	Attempts in understanding of and response to the questions. The steps in the proof / computation are basically correct, but involve some major flaws.	Poor understanding of and response to the subject. The steps in the proof / computation are largely incorrect.
Use of mathematical terminology and notation, Style and Clarity of language (20 % weighting)	Notation is skillfully used; terminology is used flawlessly. The steps in the proof / computation are well-written, easy to follow, and approaches elegance. For the result, give needed appropriate interpretation Contains no (or very few) mechanical, spelling, and/or grammatical errors	Notation and terminology are used correctly with only a few minor flaws. The steps in the proof / computation are mostly easy to follow, but may contain unnecessary detail or appear awkward. For the result, give some appropriate interpretation includes few mechanical, spelling, and/or grammatical errors	Notation and terminology are used correctly with some moderate errors. The steps in the proof / computation are basically understandable except some parts are not very clear. For the result, give few appropriate interpretation contains some mechanical, spelling, and/or grammatical errors	Notation and terminology are used correctly with some major errors. The proof/computation ends abruptly, and it may not be clear whether the writer knows how to assemble the parts of the argument. For the result, give no interpretation contains many mechanical, spelling, and/or grammatical errors	Terminology and notation are incorrectly and inconsistently used. The proof/computation is disorganized and confused. No interpretation for results. contains incomplete thoughts and confusing mechanical, spelling, and/or grammatical errors

CR Assessment – Final Examination

	Excellent 10/9	Good 8/7/6	Satisfactory 5/4	Marginal Pass 3/2/1	Fail 0
Understanding of & Response to Subject Matter (20%)	Excellent understanding of and response to the subject, demonstrates a very high general level of cognizance of, awareness of and insight into the subject.	Good understanding of and response to the subject, demonstrates a high general level of cognizance of, awareness of, and insight into the subject.	Satisfactory understanding of and response to the subject, demonstrates an adequate general level of cognizance of, awareness of, and insight into the subject.	Attempts in understanding of and response to the subject, demonstrates a limited and only rudimentary level of cognizance of, awareness of, and insight into the subject.	Poor understanding of and response to the subject, demonstrates a markedly inadequate and substandard general level of cognizance of, awareness of, and insight into the subject.
Computational & Analytical &Ability (70%)	Excellent level of analytical ability, demonstrates a very high level of conceptual understanding and ability to make sophisticated, coherent and warranted conceptual distinctions.	Good level of analytical ability, demonstrates a high level of conceptual understanding, and ability to make sophisticated, coherent and warranted conceptual distinctions.	Satisfactory level of analytical ability, demonstrates an adequate level of conceptual understanding, and ability to make sophisticated, coherent and warranted conceptual distinctions.	Attempts in exercising analytical ability, demonstrates a limited and only rudimentary level of conceptual understanding, and ability to make sophisticated, coherent and warranted conceptual distinctions.	Poor level of analytical ability, demonstrates a markedly inadequate and substandard level of conceptual understanding, and ability to make sophisticated, coherent and warranted conceptual distinctions.
Language, Diction and Grammar/ Overall impression (10 %)	Excellent level of use of language and diction, demonstrates a very high level of the choice and use of grammar, words and phrases, vocabulary, fluency, terminology, idiom, style, or expression.	Good level of use of language and diction, demonstrates a high level of the choice and use of grammar, words and phrases, vocabulary, fluency, terminology, idiom, style, or expression.	Satisfactory level of use of language and diction, demonstrates an adequate level of the choice and use of grammar, words and phrases, vocabulary, fluency, terminology, idiom, style, or expression.	Attempts in the use of language and diction, demonstrates a limited and only rudimentary level of the choice and use of grammar, words and phrases, vocabulary, fluency, terminology, idiom, style, or expression.	Poor level of use of language and diction, demonstrates an inadequate and substandard level of the choice and use of grammar, words and phrases, vocabulary, fluency, terminology, idiom, or style.

Grading System

Letter Grade	Academic Performance	Grade Point
A	Excellent	4.00
A-		3.67
B+	Good	3.33
B		3.00
B-		2.67
C+	Satisfactory	2.33
C		2.00
C-		1.67
D	Marginal Pass	1.00
F	Fail	0.00

Some notices on this Course

- Assignments must be handed in before the deadline. After the deadline, we refuse to accept your assignments!
- For quizzes, the mid-term test and final examination, you can not bring anything except some stationeries and water!

Mobile phones are not allowed.

- For the final examination, we can not tell you the score before the AR inform the official results. If you have any question on the score, you can make an appeal via AR.

Recommended Readings

CALCULUS

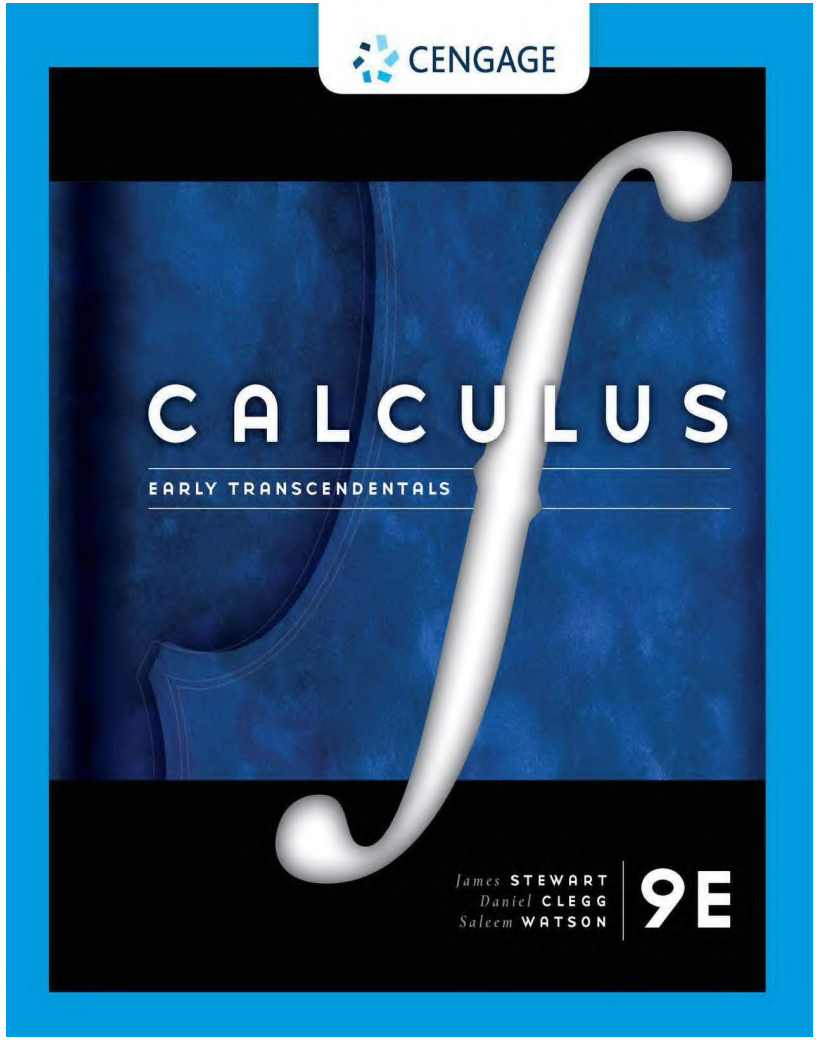
EARLY TRANSCENDENTALS

NINTH EDITION

JAMES STEWART

DANIEL CLEGG

SALEEM WATSON



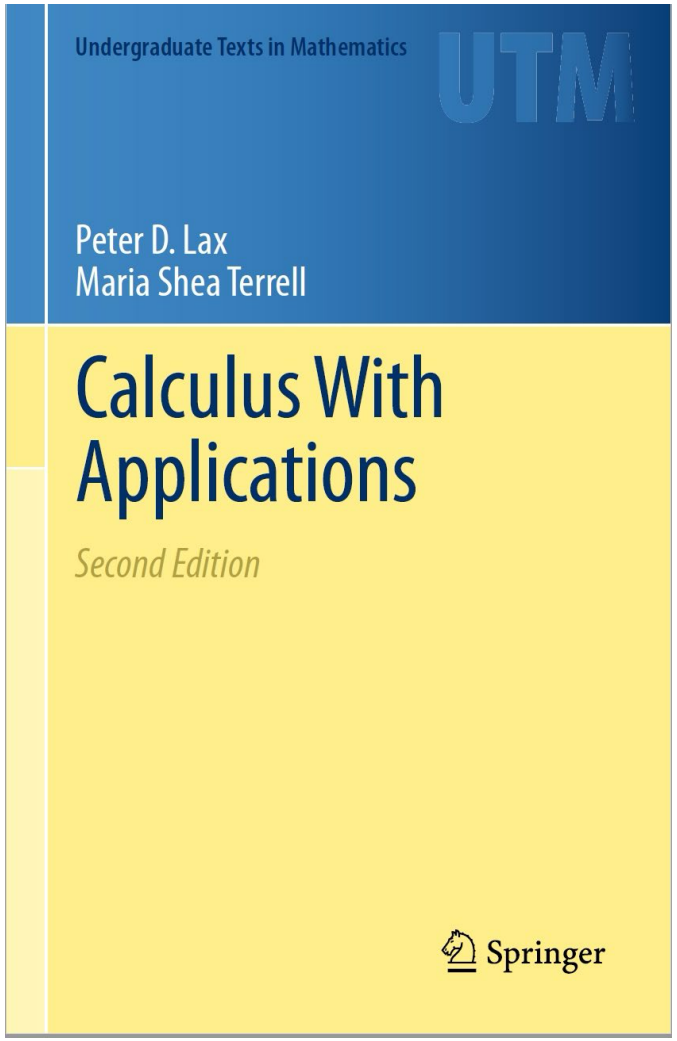
Recommended Readings

Calculus with Applications

Second Edition

Peter D. Lax

Maria Shea Terrell



Recommended Readings

CALCULUS

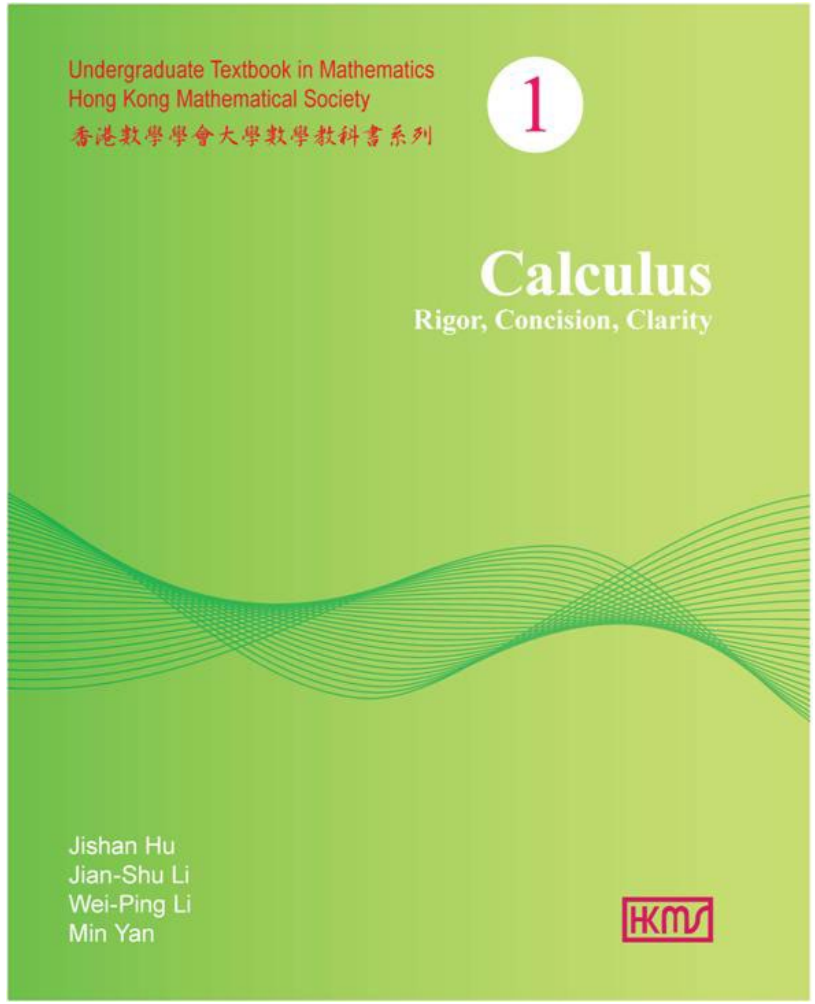
Rigor, Concision, Clarity

Jishan Hu

Jian-Shu Li

Wei-Ping Li

Min Yan



Available at:

https://www.math.hkust.edu.hk/~majhu/Math023/Notes_2007.pdf

Course Content

- Chapter 1 Numbers and Sequences
- Chapter 2 Functions and Continuity
- Chapter 3 Derivatives
- Chapter 4 Applications of Differentiation
- Chapter 5 Antiderivatives
- Chapter 6 The Riemann Integral